

#8.1.3,

Let R be a Euclidean domain. Let $m = \min \{N(a) \mid a \in R, a \neq 0\}$.

Prove that every non-zero element of R of norm m is a unit.

Proof. Let a non-zero element $a \in R$ be given with $N(a) = m$.

Now, by the Euclidean Algorithm,

$$1 = q \cdot a + r \text{ for some } q, r \in R \text{ with } r = 0 \text{ or } N(r) < N(a).$$

By the minimality, $r = 0$, so $1 = q \cdot a$. Thus a is unit. $\square$

#8.1.4,

Let R be a Euclidean domain.

a). If $(a, b) = 1$ and $a \mid bc$, then $a \mid c$.

If $a \mid bc$ where a, b are nonzero, then $a \cdot (a, b)^{-1}$ divides c .

Proof. Suppose that $a \nmid c$ under $(a, b) = 1$ and $a \mid bc$.

Since $a \mid c \Leftrightarrow (c) \subseteq (a)$, $a \nmid c \Leftrightarrow (c) \not\subseteq (a)$. And, since $a \mid bc$, $(bc) \subseteq (a)$.

$$(a, b) = (d)$$

